Waveshare ESP32-S3-Touch-LCD-2: Precision Current Monitor & Logger

A professional-grade, high-visibility directional current monitoring instrument and data logger built for the Waveshare ESP32-S3-Touch-LCD-2 development board and the INA219 sensor.

🚀 Key Features

📊 Advanced Dashboard

- **Calibrated Analogue Gauge**: A central circular scale (0-2000mA) with a smooth-motion indicator arc.
- **Directional Color Coding**:
  - 🟢 **Green**: Charging / Power Source (> 1mA).
  - 🔴 **Red**: Discharging / Power Load (< -1mA).
  - ⚪ **White**: Neutral Deadband (-1mA to 1mA) for visual stability.
- **High-Visibility Readout**: Uses **Montserrat 48** bold font for the central numeric display and **Montserrat 24** for units.
- **Auto-Scaling Units**: Automatically switches from **mA** (integer) to **Amps** (2 decimals) when current exceeds 999mA.

💾 Data Logging System

- **On-Demand Setup**: Prompts for Date & Time setup only when logging is first initiated, ensuring meaningful timestamps.
- **Manual Control**: Dedicated **START/STOP** touch button to control logging sessions.
- **Reliable Timestamps**: Uses a hardware-backed `millis()` offset to ensure precisely incrementing records even without an active internet connection.
- **Automatic File Rotation**: Generates CSV files named by timestamp and automatically starts a new file every **5,000 records**.
- **Configurable Intervals**: Integrated **Settings Menu** (Gear icon) to change logging frequency from **1 second** to **10 minutes**.

🛠️ Hardware Configuration

Pin Mapping

Component	Function	Pins
LCD (ST7789)	SPI (HSPI)	CS: 45, DC: 42, RST: 0, MOSI: 38, SCK: 39, BL: 1
SD Card	SPI (Shared)	CS: 41, MISO: 40, MOSI: 38, SCK: 39
I2C Bus	Shared	SDA: 48, SCL: 47
Touch (CST816D)	I2C	INT: 46, RST: 0
INA219 Sensor	I2C	Address: 0x40 (Default)

Board Settings (Arduino IDE)

- **Board**: `ESP32S3 Dev Module`
- **USB CDC On Boot**: `Enabled` (Required for Serial Debugging)
- **Flash Size**: `16MB`
- **PSRAM**: `OPI PSRAM`

📦 Software Requirements

The following libraries must be installed via the Arduino Library Manager:

- `lvgl` (v9.x)
- `Adafruit_ST7789`
- `Adafruit_GFX`
- `Adafruit_INA219`
- `CST816S` (by fbiego)

LVGL Configuration

Ensure your `lv_conf.h` has the following fonts enabled:

```
#define LV_FONT_MONTSERRAT_14 1
#define LV_FONT_MONTSERRAT_24 1
#define LV_FONT_MONTSERRAT_34 1
#define LV_FONT_MONTSERRAT_48 1
```

📖 Usage

1. **Boot**: The device starts directly into the real-time monitoring dashboard.

2. **Monitor**: View the live current flow on the analogue gauge and digital readout.
3. **Log**: Press the green **START** button. If the time is not set, use the rollers to select the current Date/Time and press **START LOGGING**.
4. **Configure**: Press the **Gear Icon** in the top right to adjust the logging frequency.
5. **Retrieve Data**: Insert the SD card into a computer to access the `.csv` log files.